

Step 2: Data types

Data flow

Data

Analyses

Create model

Data Quality And Insights Report



Data quality and insights report: Bank Marketing Data Quality Report

Target column	Type	Dataset	Date
y	Classification	bank-marketing.csv	26 de mayo de 2025, 16:03 CEST



Summary

Dataset statistics

Key	Value	
Number of features	21	
Number of rows	41188	
Missing	0%	
Valid	100%	
Duplicate rows	0.0583%	

Feature type	Count	
numeric	10	
categorical	9	
text	0	
datetime	0	
binary	1	
unknown	0	

High Priority Warnings

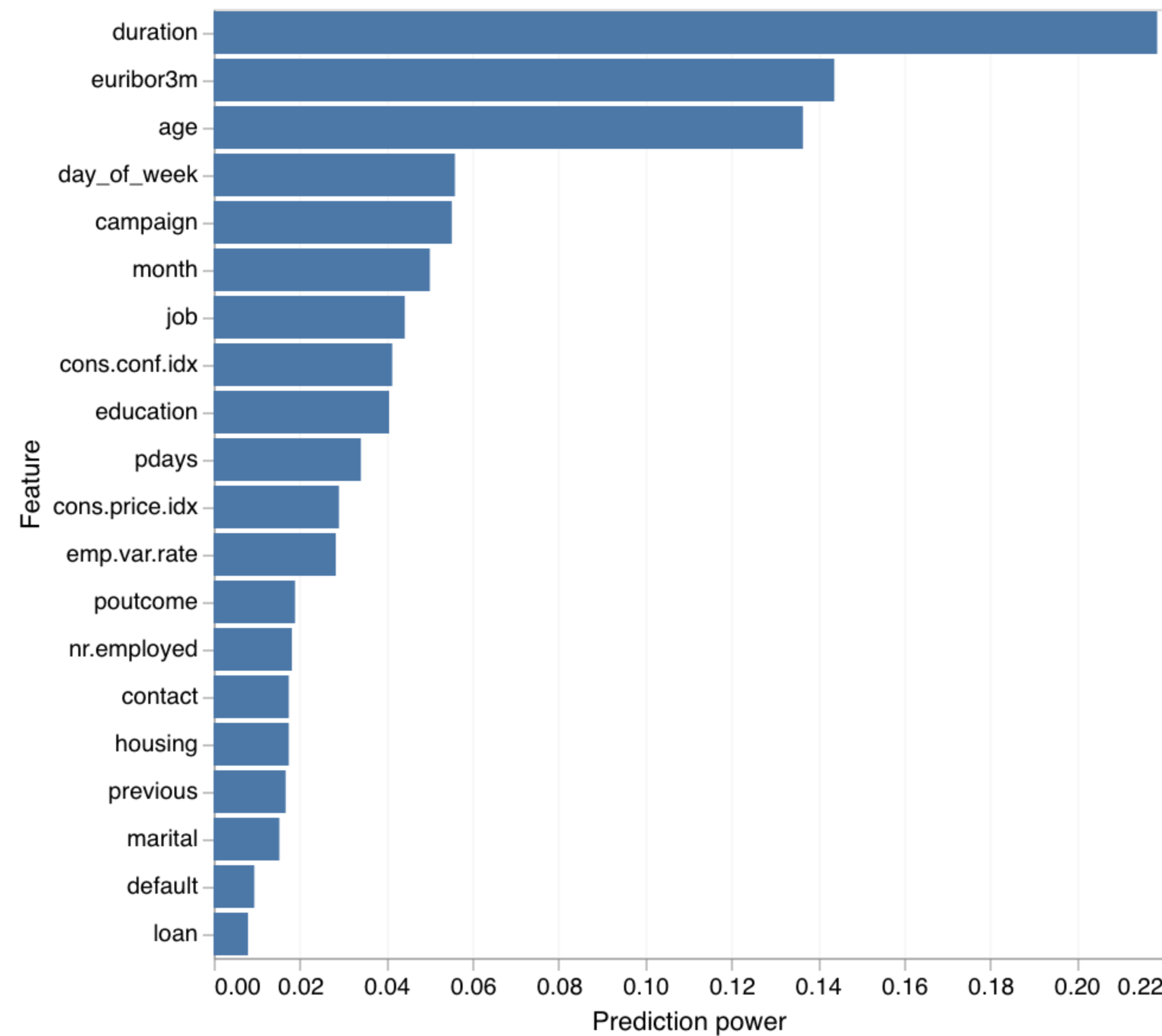
No high severity warnings were detected in the data.

Feature summary

See a summary of the features ordered by the prediction power. Prediction power is measured by stratified splitting the data into 80%/20% training and validation folds. We fit a model for each feature separately on the training fold after applying minimal feature pre-processing and measure prediction performance on the validation data.

- The scores are normalized to the range [0,1].
- Higher prediction power scores, toward 1, indicate columns that are more useful for predicting the target on their own.
- Lower scores, toward 0 point to columns that contain little useful information for predicting the target on their own. Although it can happen that a column is uninformative on its own but is useful in predicting the target when used in tandem with other features, a low score usually indicates the feature is redundant.
- A score of 1 implies perfect predictive abilities, which often indicates an error called target leakage. The cause is typically a column that will not be available at prediction time such as a duplicate of the target.

Feature	Prediction power	Type	Valid	Missing	High severity warnings	Medium severity warnings	
duration	0.219	numeric	100%	0%	0	0	
euribor3m	0.144	numeric	100%	0%	0	0	
age	0.137	numeric	100%	0%	0	0	
day_of_week	0.056	categorical	100%	0%	0	0	
campaign	0.0552	numeric	100%	0%	0	0	
month	0.0501	categorical	100%	0%	0	1	



Prediction power of the features. Higher prediction power scores, toward 1, indicate columns that are more useful for predicting the target on their own. Lower scores, toward 0 point to columns that contain little useful information for predicting the target on their own.

Duplicate rows

Canvas detected that 0.0583% of the data are duplicate. Some data sources could include valid duplicates. Other data sources could have duplicates that point to problems in data collection. Duplicate samples that result from faulty data collection could interfere with machine learning processes that rely on splitting the data into independent training and validation folds.

The following are examples of processes that can suffer from duplicated samples:

- Quick model analysis
- Prediction power estimation
- Hyper-parameters optimization

The most common duplicate rows are presented below. The number of occurrences of a row is given in left most column named Duplicate count. You can remove duplicate samples from the dataset using the Drop duplicates transform under Manage rows.

Note: features of type vector are ignored in duplicate rows detection

Duplicate count	age	job	marital	education	default	housing	loan
2	24	services	single	high.school	no	yes	no
2	27	technician	single	professional.course	no	no	no
2	32	technician	single	professional.course	no	yes	no
2	35	admin.	married	university.degree	no	yes	no
2	36	retired	married	unknown	no	no	no
2	39	admin.	married	university.degree	no	no	no

Duplicate rows Medium

i We found that 0.0583% of the data are duplicate. Some data sources could include valid duplicates and in other cases these duplicates could point to problems in data collection. Duplicate samples resulting from faulty data collection, could derail machine learning processes that rely on splitting to independent training and validation folds. For example quick model scores, prediction power estimation and automatic hyper parameter tuning. Duplicate samples could be removed from the dataset using the Drop duplicates transform under Manage rows.

Anomalous samples

Canvas detects anomalous samples using the Isolation forest algorithm after basic preprocessing. The isolation forest associates an anomaly score to each sample (row) of the dataset.

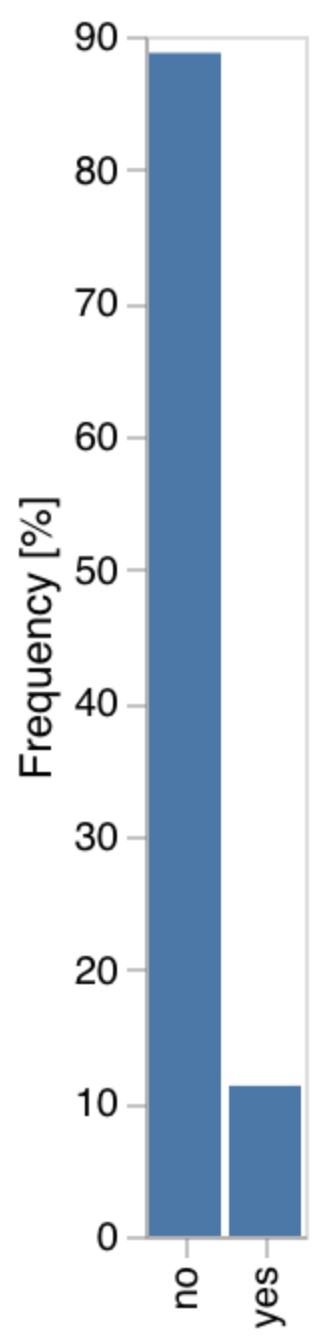
- Low anomaly scores indicate anomalous samples.
- High scores are associated with non-anomalous samples.
- Samples with negative anomaly score are usually considered anomalous and samples with positive anomaly score are considered non-anomalous.

When you look at a sample that might be anomalous, we recommend that you pay attention to unusual values. For example, you might have anomalous values that result from errors in gathering and processing the data. The following is an example of the most anomalous samples according to the Canvas's implementation of the isolation forest algorithm. We recommend using domain knowledge and business logic when you examine the anomalous samples.

Anomaly scores	age	job	marital	education	default	housing	loan
-0.2	86	retired	married	unknown	unknown	yes	yes
-0.187	45	unemployed	married	professional.course	unknown	no	no
-0.184	66	retired	married	basic.4y	unknown	unknown	unkr
-0.184	20	student	single	unknown	no	yes	yes
-0.183	73	retired	divorced	high.school	no	no	yes
-0.182	75	housemaid	divorced	basic.4y	no	no	no

Target column

key	value	
Number of classes	2	
Valid	100%	
Missing	0%	



Histogram of the frequent values of the target column.

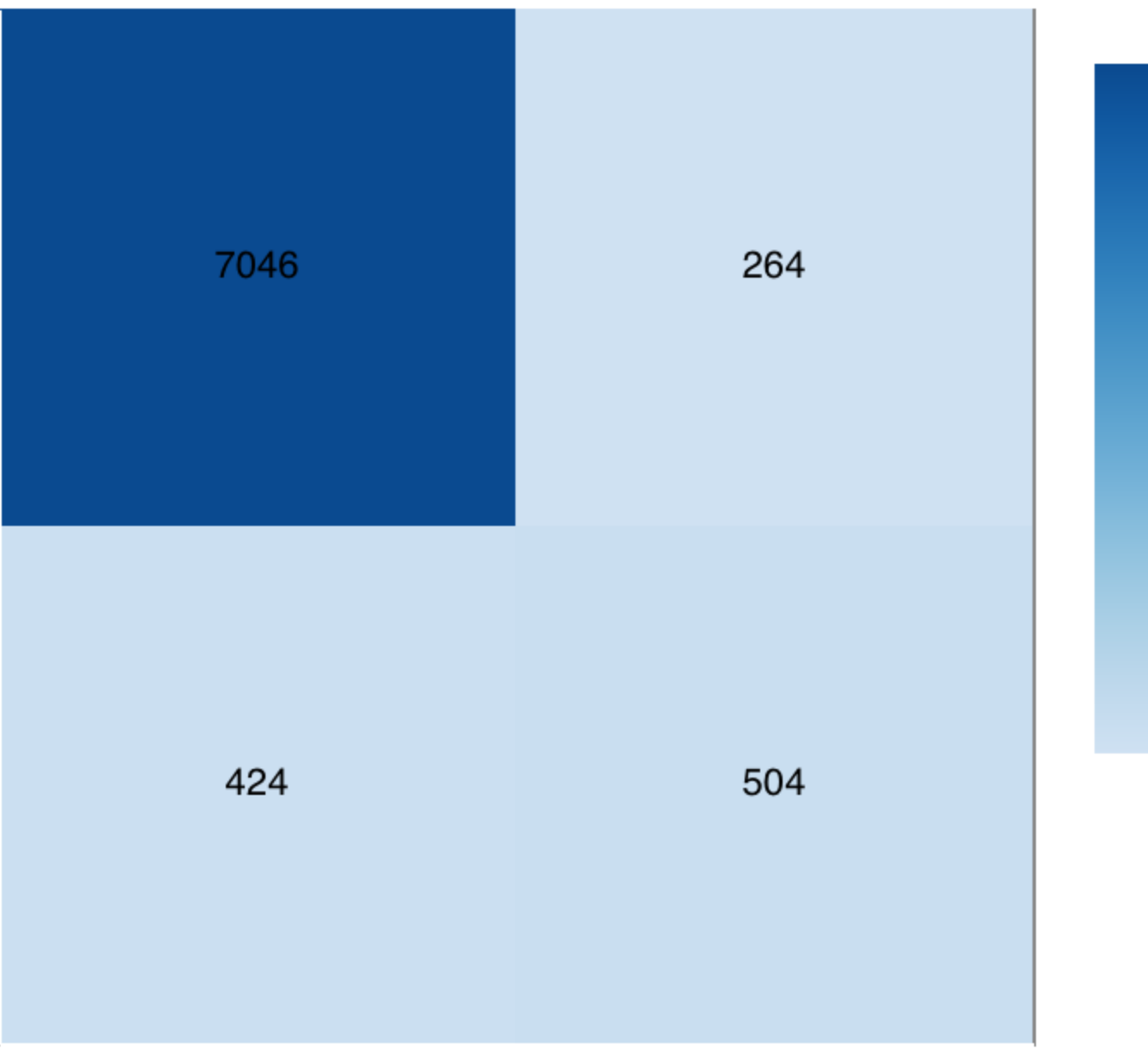
Quick model

Quick model provides an estimate of the expected predicted quality of a model that you train on your data.

The data is split into training and validation folds where Canvas uses 80% of the samples for training and 20% of the values for validation. For classification the sample is stratified split. For a stratified split, each data partition has the same ratio of labels. For classification problems, it's important to have the same ratio of labels between the training and classification folds. Canvas trains the XGBoost model with the default hyper-parameters. It applies early stopping on the validation data and performs minimal feature pre-processing.

Metric	Validation scores	Train scores	
Accuracy	0.916	0.932	
Balanced accuracy	0.753	0.793	
ROC-AUC	0.949	0.964	
F1	0.594	0.669	
Precision	0.656	0.736	
Recall	0.543	0.614	

class	precision	recall	f1-score	support	
no	0.9432396251673361	0.9638850889192886	0.9534506089309879	7310.0	
yes	0.65625	0.5431034482758621	0.5943396226415094	928.0	



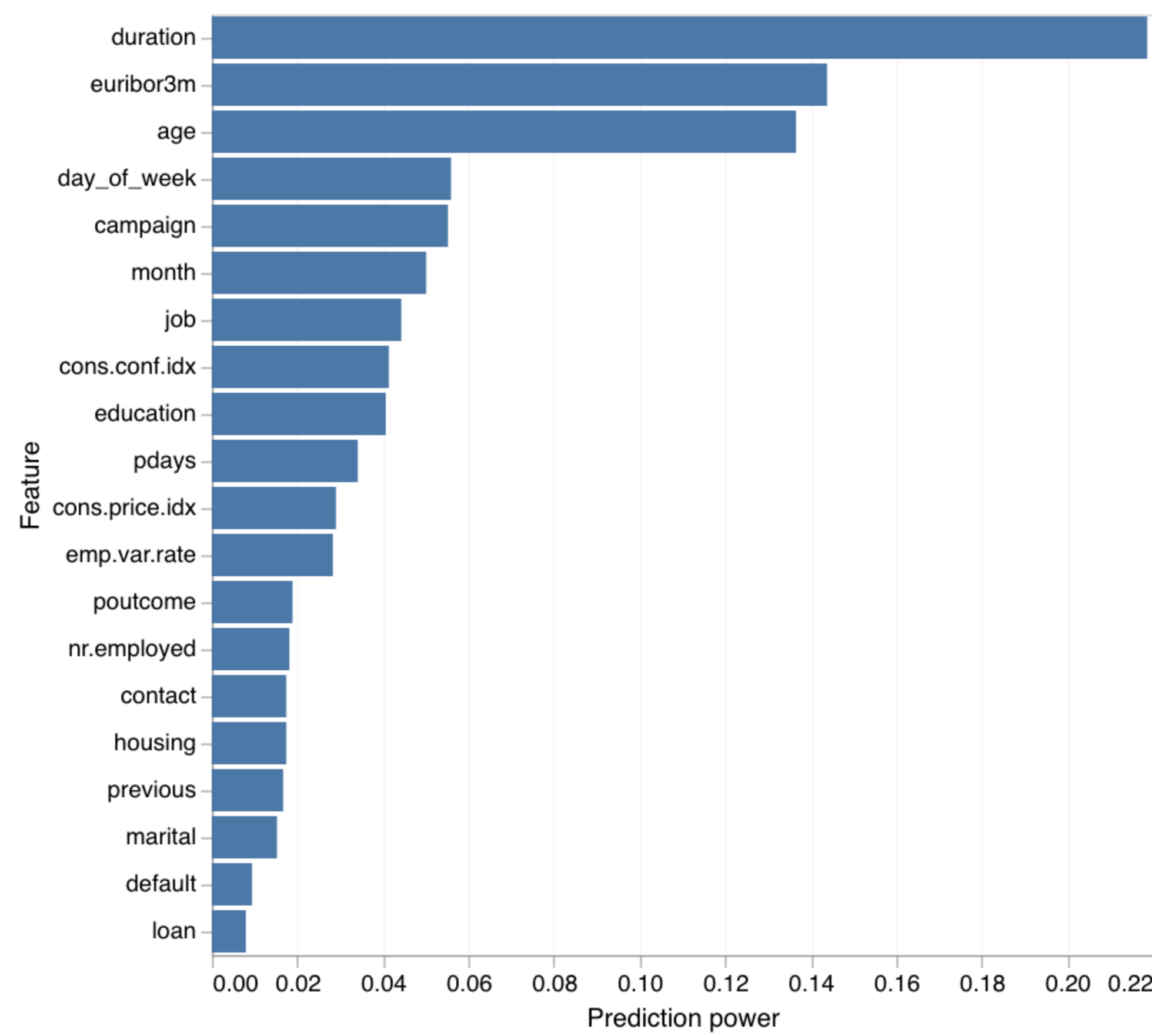
Confusion matrix for the validation fold.

Feature summary

See a summary of the features ordered by the prediction power. Prediction power is measured by stratified splitting the data into 80%/20% training and validation folds. We fit a model for each feature separately on the training fold after applying minimal feature pre-processing and measure prediction performance on the validation data.

- The scores are normalized to the range [0,1].
- Higher prediction power scores, toward 1, indicate columns that are more useful for predicting the target on their own.
- Lower scores, toward 0 point to columns that contain little useful information for predicting the target on their own. Although it can happen that a column is uninformative on its own but is useful in predicting the target when used in tandem with other features, a low score usually indicates the feature is redundant.
- A score of 1 implies perfect predictive abilities, which often indicates an error called target leakage. The cause is typically a column that will not be available at prediction time such as a duplicate of the target.

Feature	Prediction power	Type	Valid	Missing	High severity warnings	Medium severity warnings	
duration	0.219	numeric	100%	0%	0	0	
euribor3m	0.144	numeric	100%	0%	0	0	
age	0.137	numeric	100%	0%	0	0	
day_of_week	0.056	categorical	100%	0%	0	0	
campaign	0.0552	numeric	100%	0%	0	0	
month	0.0501	categorical	100%	0%	0	1	

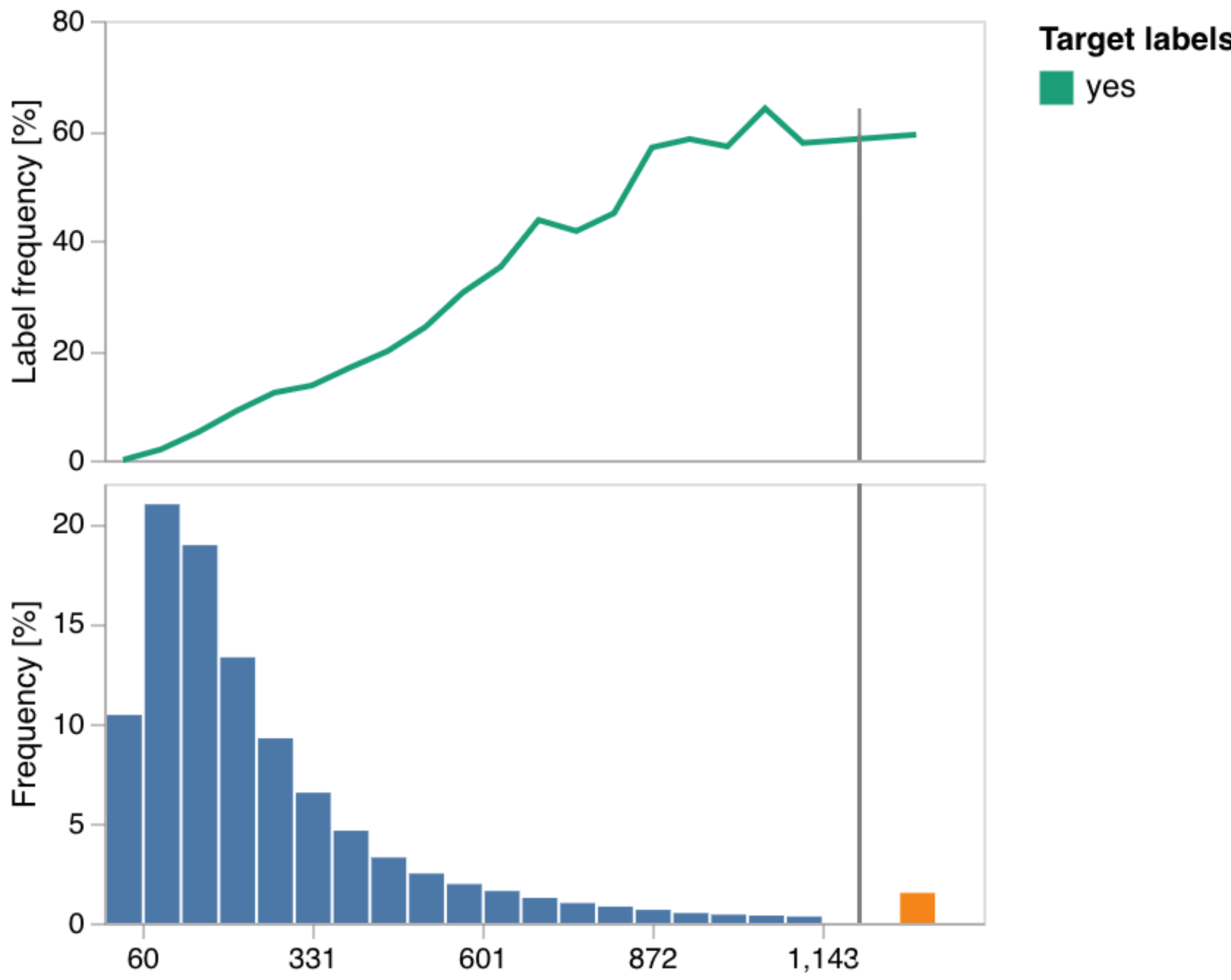


Prediction power of the features. Higher prediction power scores, toward 1, indicate columns that are more useful for predicting the target on their own. Lower scores, toward 0 point to columns that contain little useful information for predicting the target on their own.

Feature details

duration · numeric

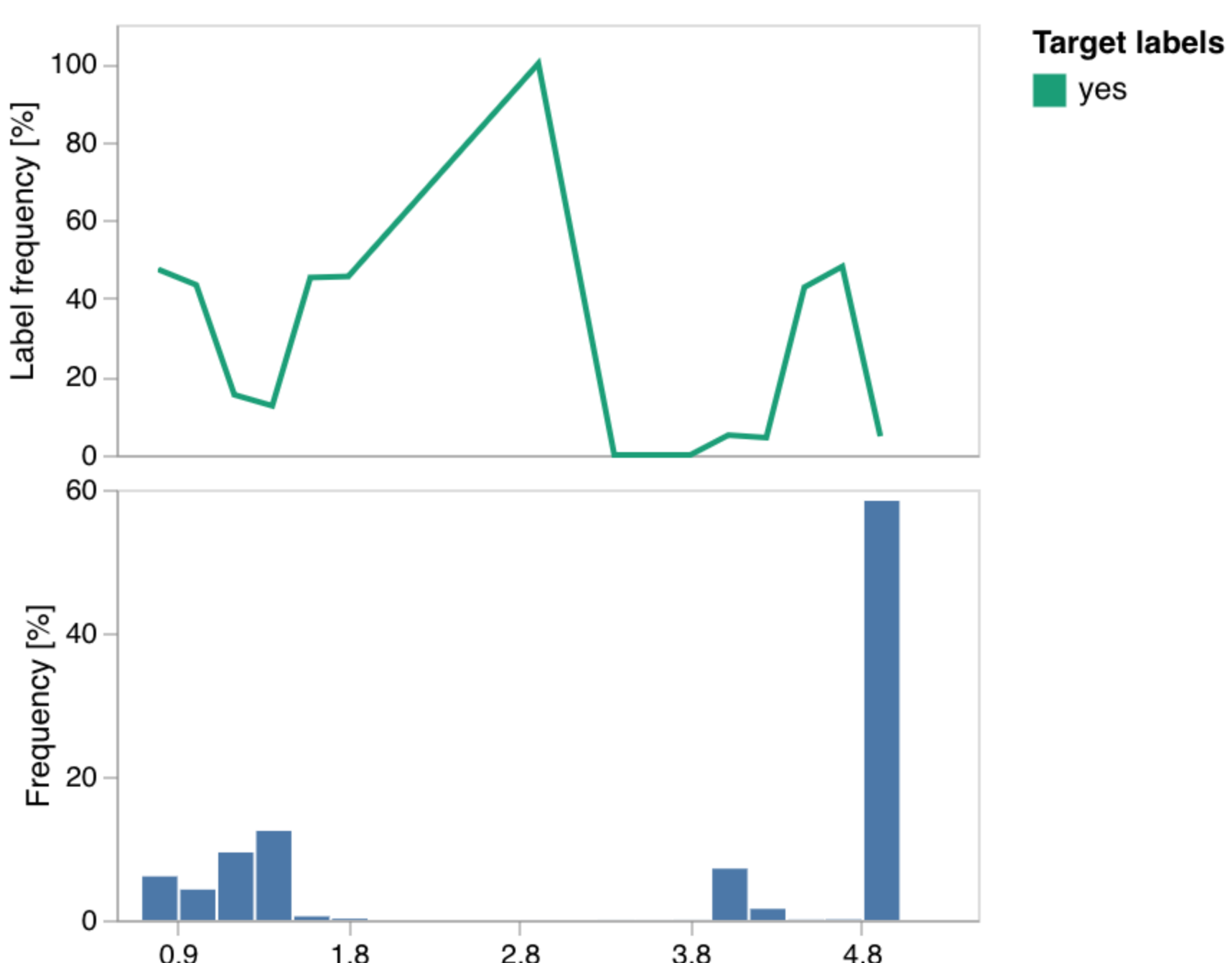
key	value	
Feature name	duration	
Type	numeric	
Prediction power	0.219	
Valid	100%	
Missing	0%	
Outliers	1.5%	
Min	0	
Max	4.92e+03	
Mean	258	



Histogram of duration with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency. The red bars contain outliers and the value below them is the outliers average.

euribor3m · numeric

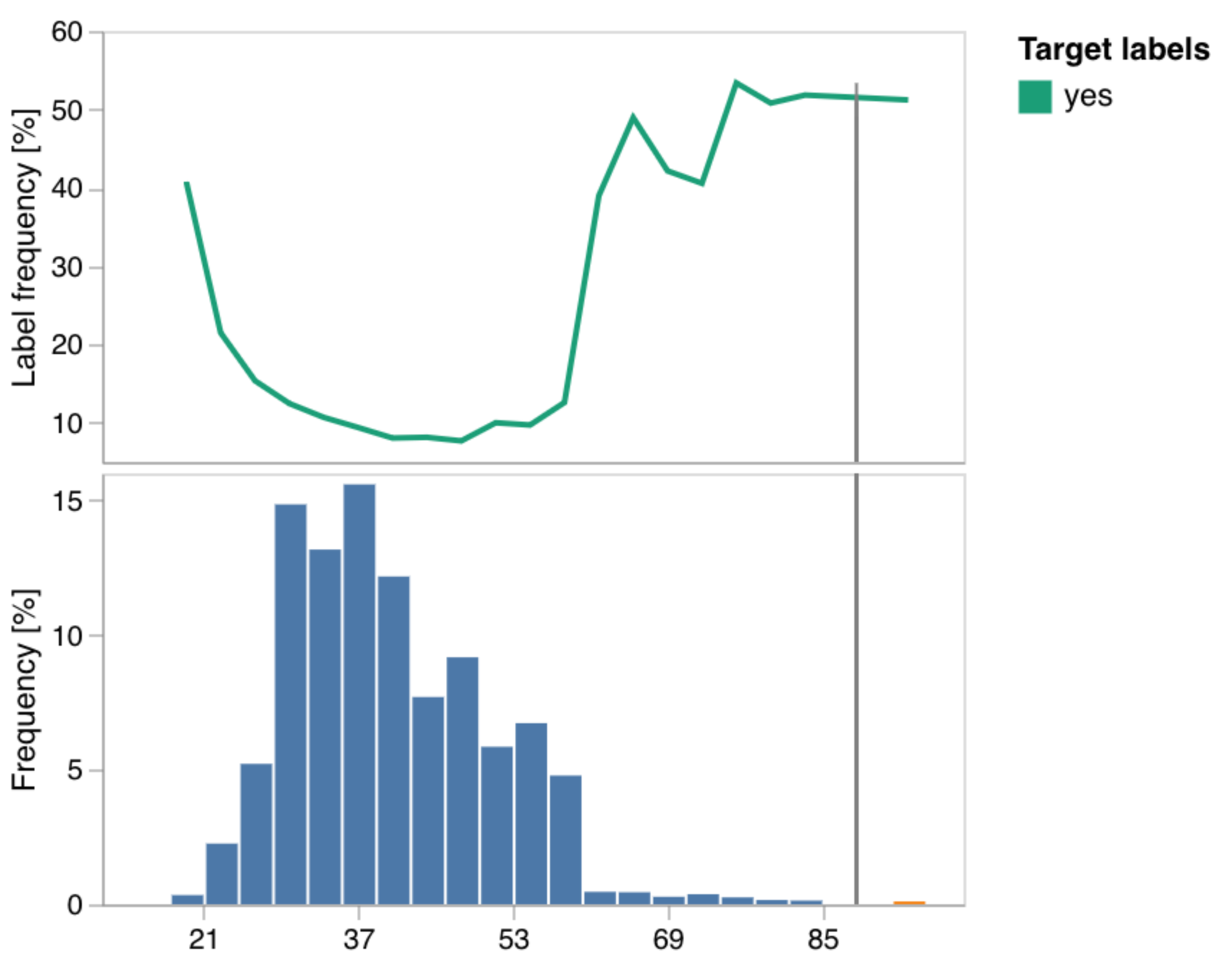
key	value	
Feature name	euribor3m	
Type	numeric	
Prediction power	0.144	
Valid	100%	
Missing	0%	
Outliers	0%	
Min	0.634	
Max	5.04	
Mean	3.62	



Histogram of euribor3m with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency.

age · numeric

key	value	
Feature name	age	
Type	numeric	
Prediction power	0.137	
Valid	100%	
Missing	0%	
Outliers	0.104%	
Min	17	
Max	98	
Mean	40	



Histogram of age with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency. The red bars contain outliers and the value below them is the outliers average.

day_of_week · categorical

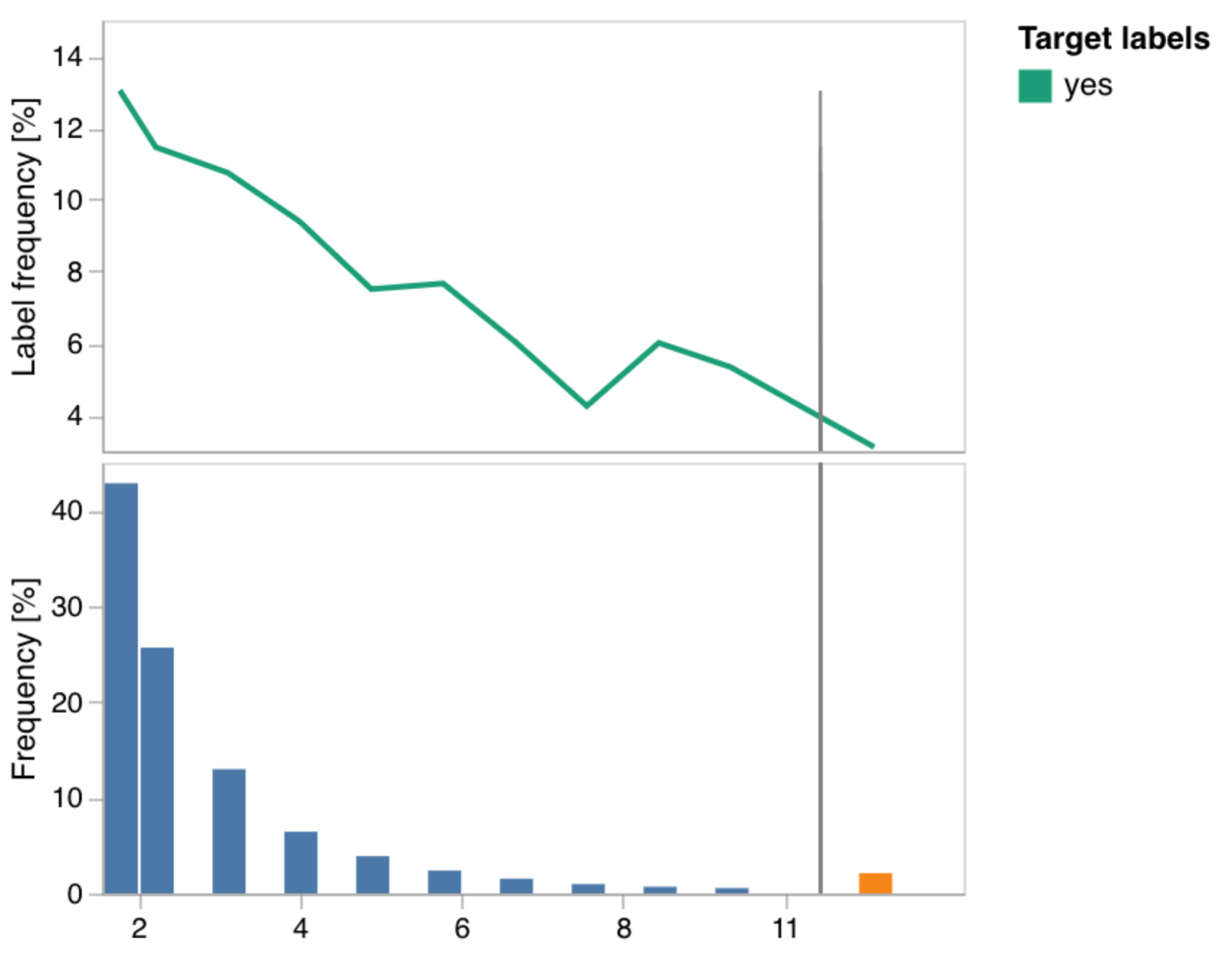
key	value	
Feature name	day_of_week	
Type	categorical	
Prediction power	0.056	
Valid	100%	
Missing	0%	



Histogram of the frequent values of day_of_week with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency.

campaign · numeric

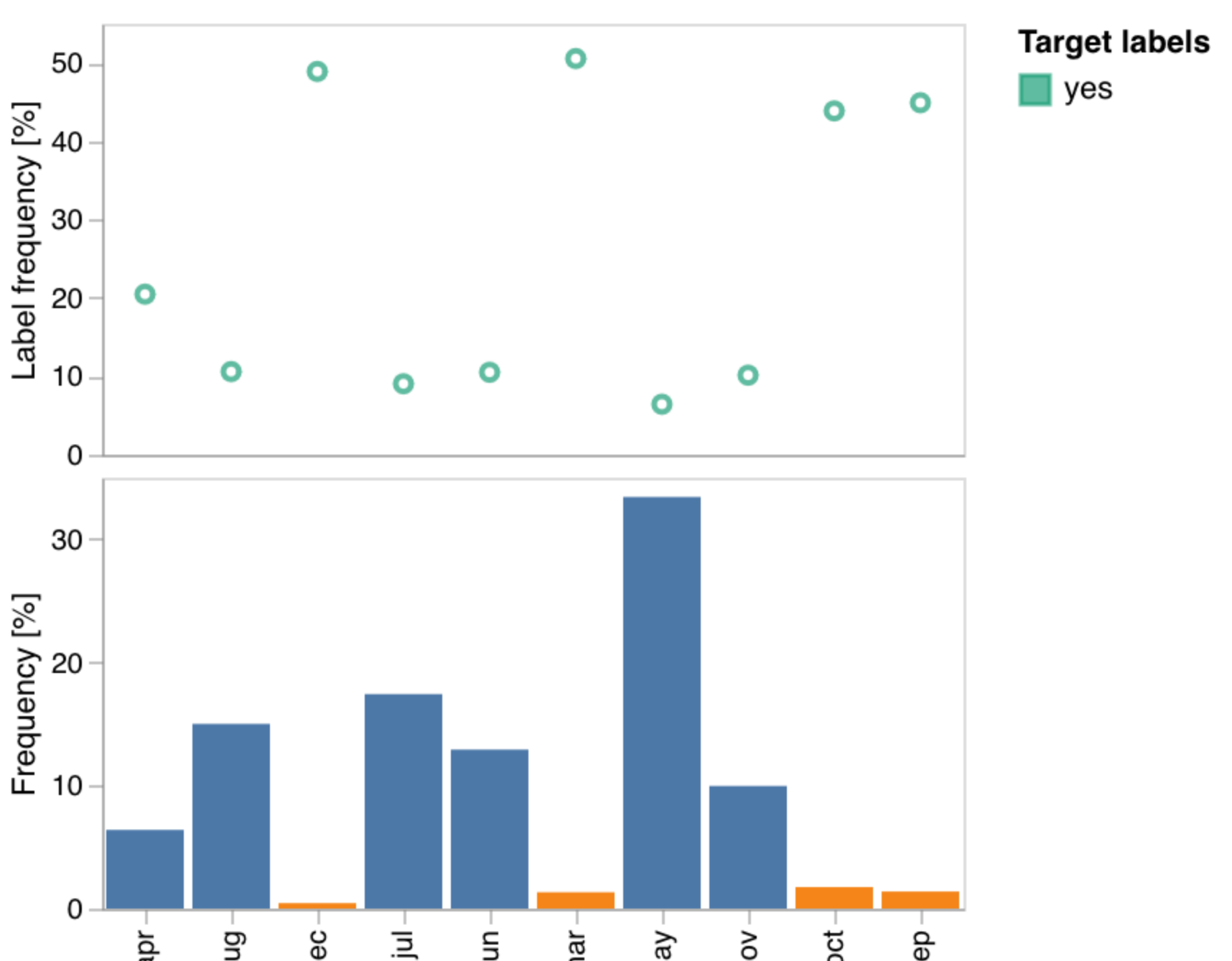
key	value	
Feature name	campaign	
Type	numeric	
Prediction power	0.0552	
Valid	100%	
Missing	0%	
Outliers	2.11%	
Min	1	
Max	56	
Mean	2.57	



Histogram of campaign with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency. The red bars contain outliers and the value below them is the outliers average.

month · categorical

key	value	
Feature name	month	
Type	categorical	
Prediction power	0.0501	
Valid	100%	
Missing	0%	



Histogram of the frequent values of month with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency.

Rare classes

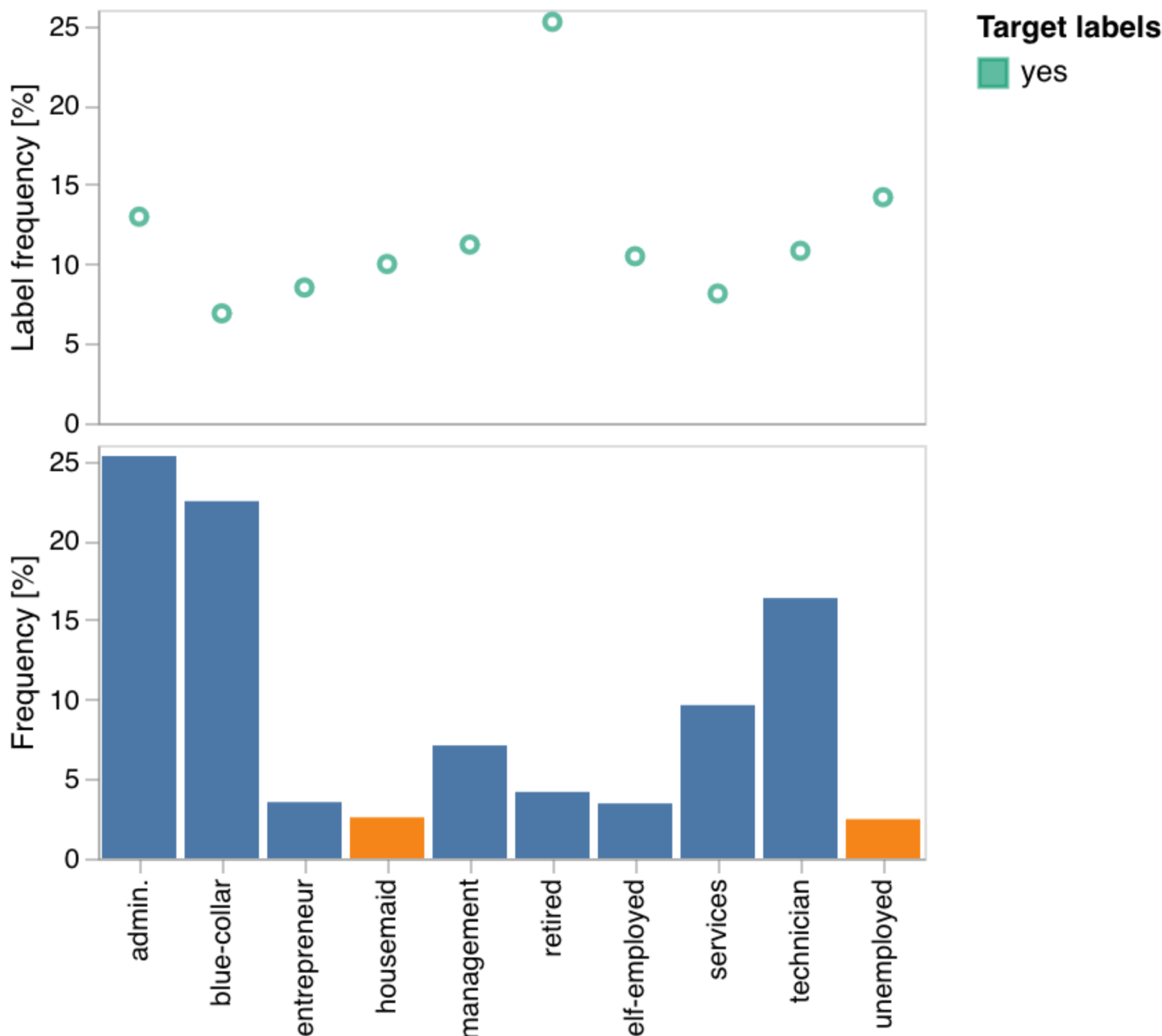
Medium

i

The feature month has 3 categories which are rare: sep, mar, dec. This could lead to overfitting and subpar prediction performance. Consider consolidating them into a single category using the Replace rare values transform under Handle outliers or Find and replace substring transform under Search and edit.

job · categorical

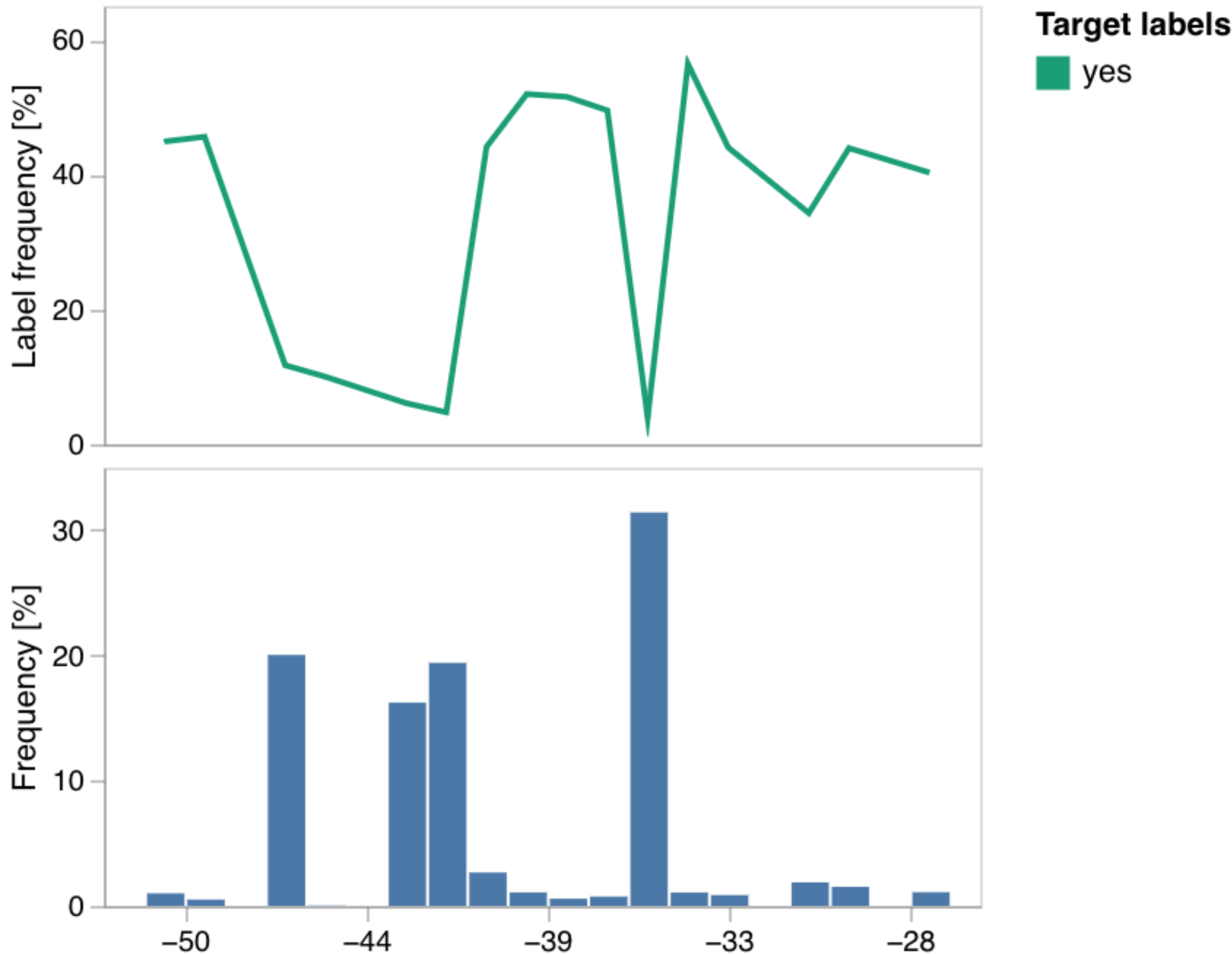
key	value	
Feature name	job	
Type	categorical	
Prediction power	0.0443	
Valid	100%	
Missing	0%	



Histogram of the frequent values of job with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency.

cons.conf.idx · numeric

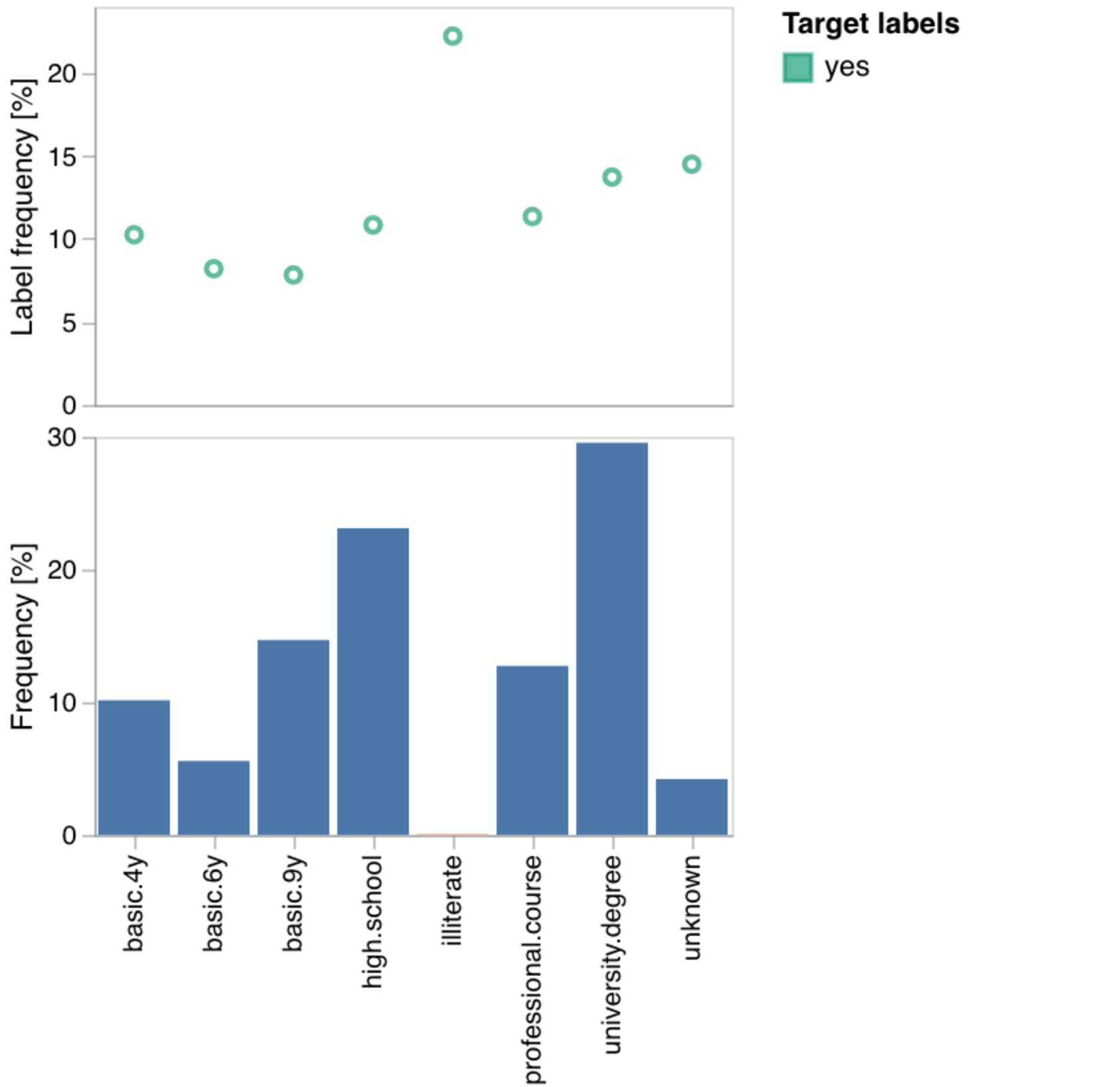
key	value	
Feature name	cons.conf.idx	
Type	numeric	
Prediction power	0.0414	
Valid	100%	
Missing	0%	
Outliers	0%	
Min	-50.8	
Max	-26.9	



Histogram of cons.conf.idx with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency.

education · categorical

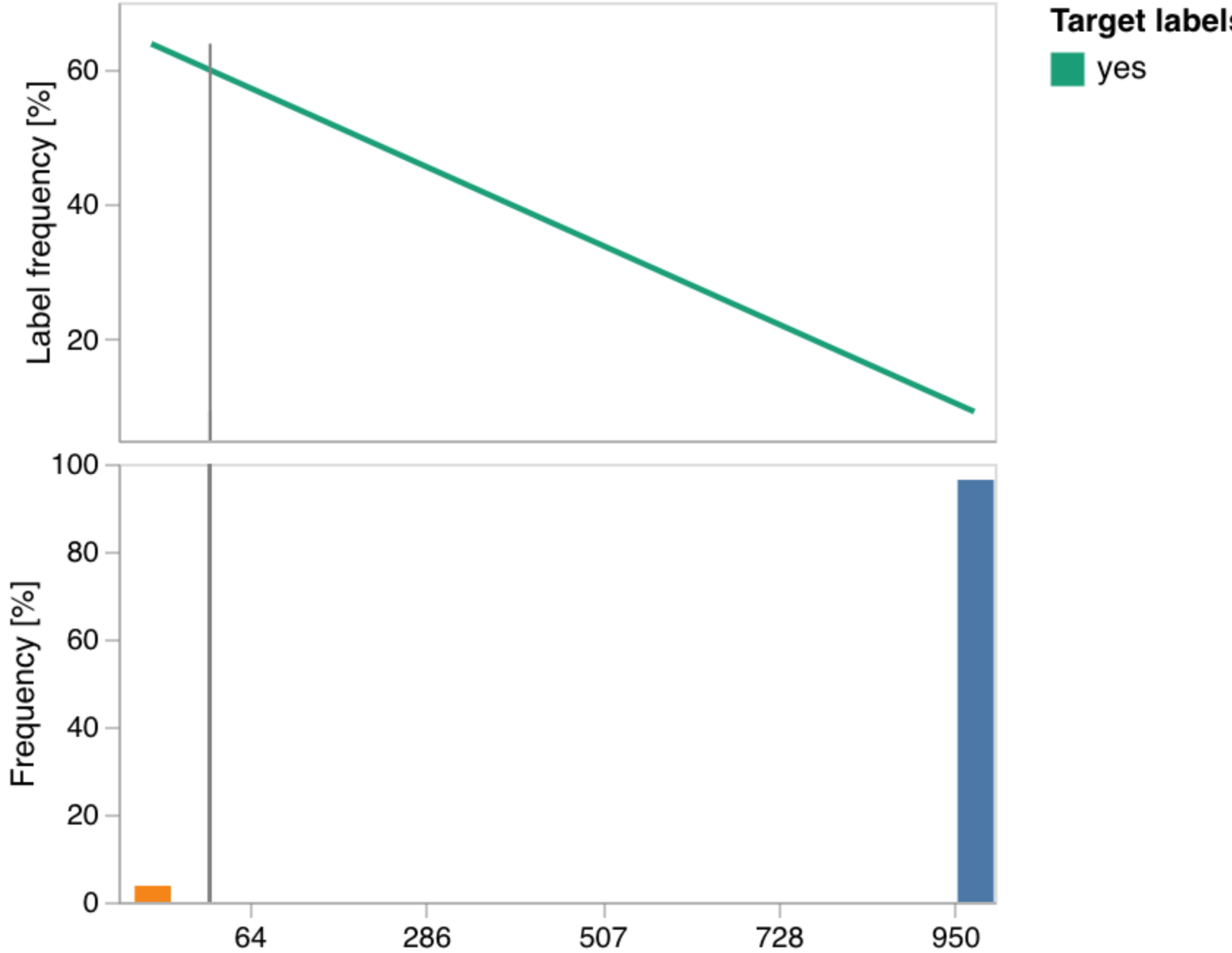
key	value	
Feature name	education	
Type	categorical	
Prediction power	0.0407	
Valid	100%	
Missing	0%	



Histogram of the frequent values of education with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency.

pdays · numeric

key	value	
Feature name	pdays	
Type	numeric	
Prediction power	0.0342	
Valid	100%	
Missing	0%	
Outliers	3.68%	
Min	0	
Max	999	
Mean	962	



Histogram of pdays with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency. The red bars contain outliers and the value below them is the outliers average.

Disguised missing value

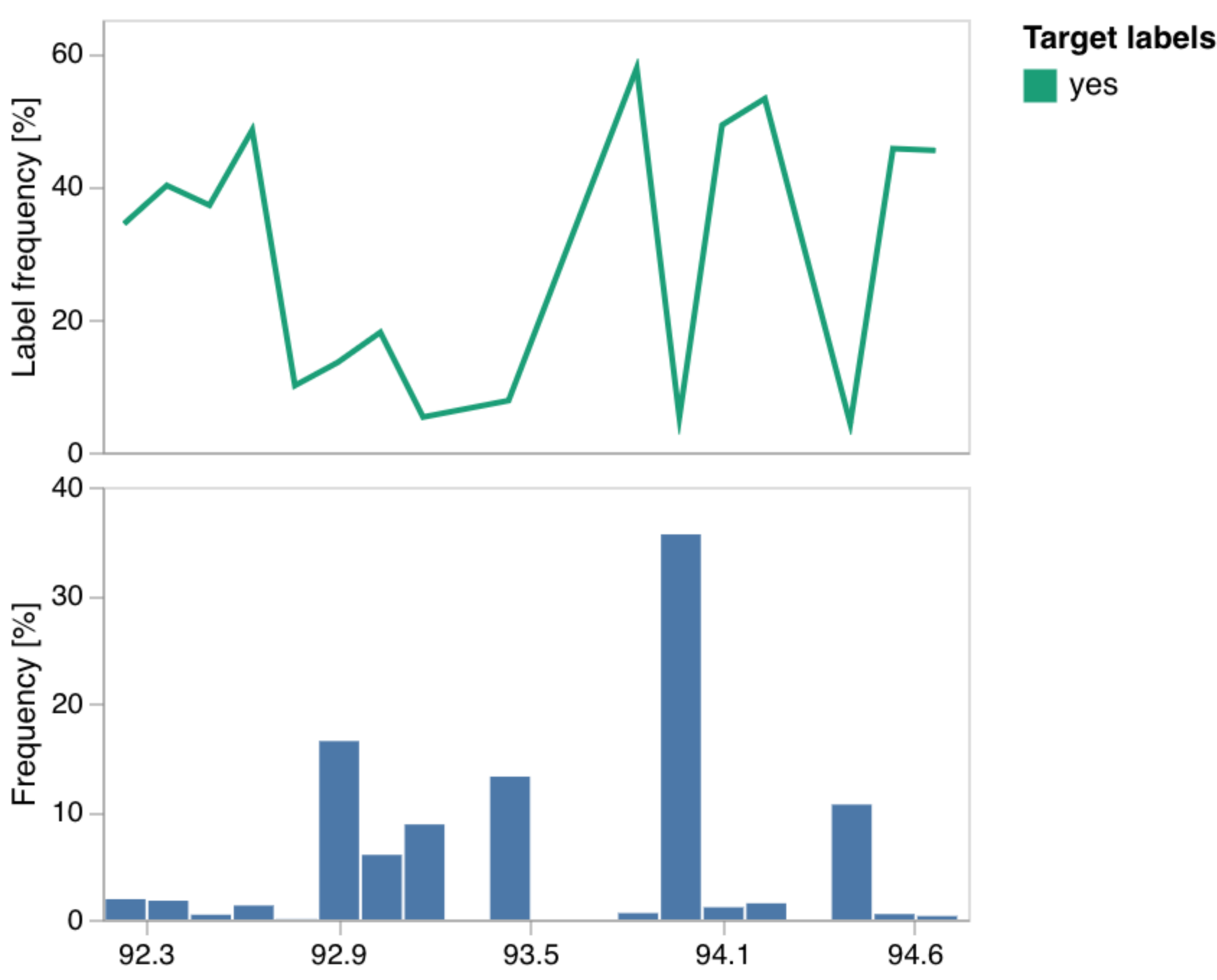
Medium

i

The frequency of the value "999" in the feature pdays is 96.3% which is uncommon for numeric features. This could point to bugs in data collection or processing. In some cases the frequent value is a default value or a placeholder to indicate a missing value. If that is the case, it is recommended to replace this value with NaNs using the Convert regex to missing transform under Search and edit.

cons.price.idx · numeric

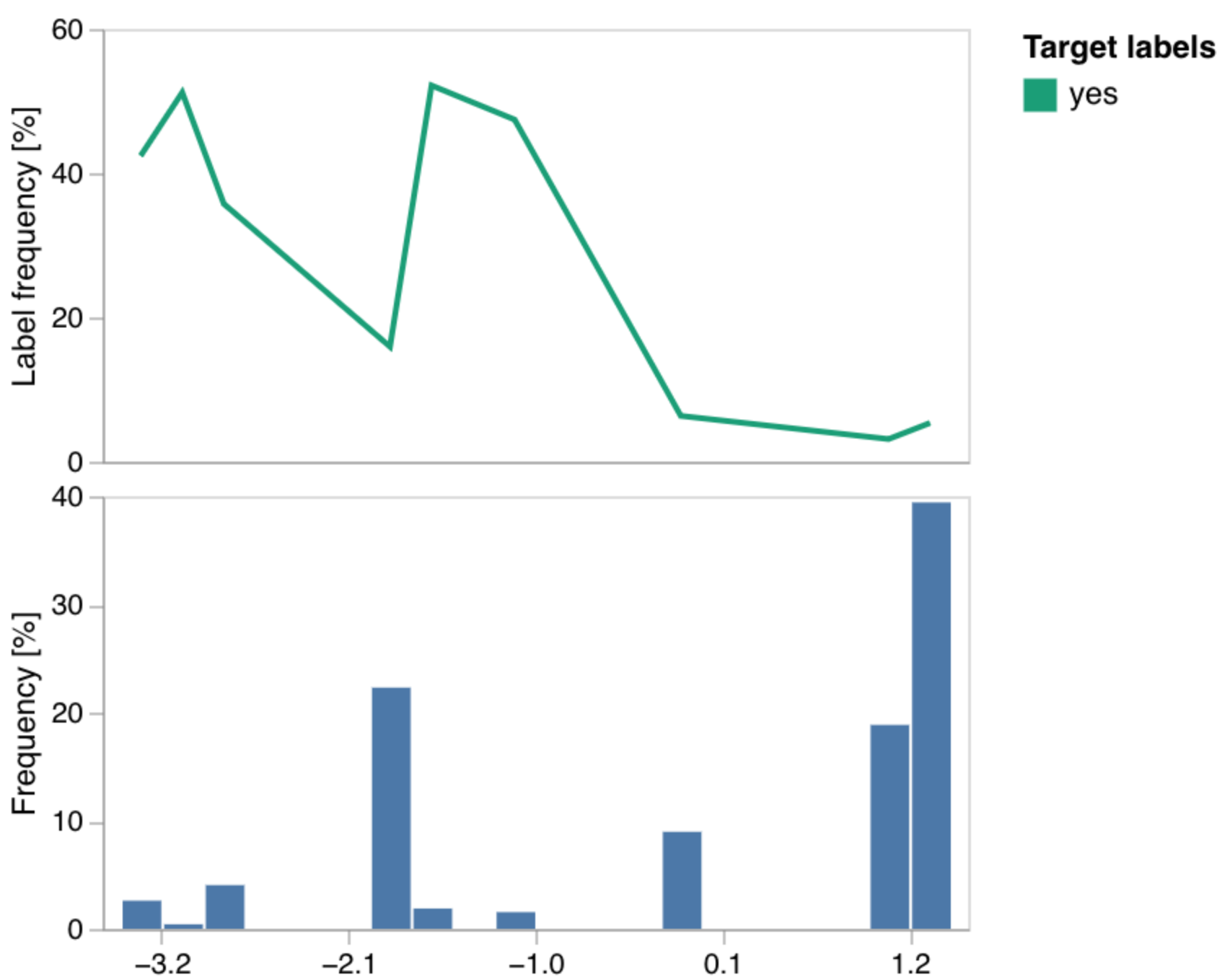
key	value	
Feature name	cons.price.idx	
Type	numeric	
Prediction power	0.0291	
Valid	100%	
Missing	0%	
Outliers	0%	
Min	92.2	
Max	94.8	
Mean	93.6	



Histogram of cons.price.idx with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency.

emp.var.rate · numeric

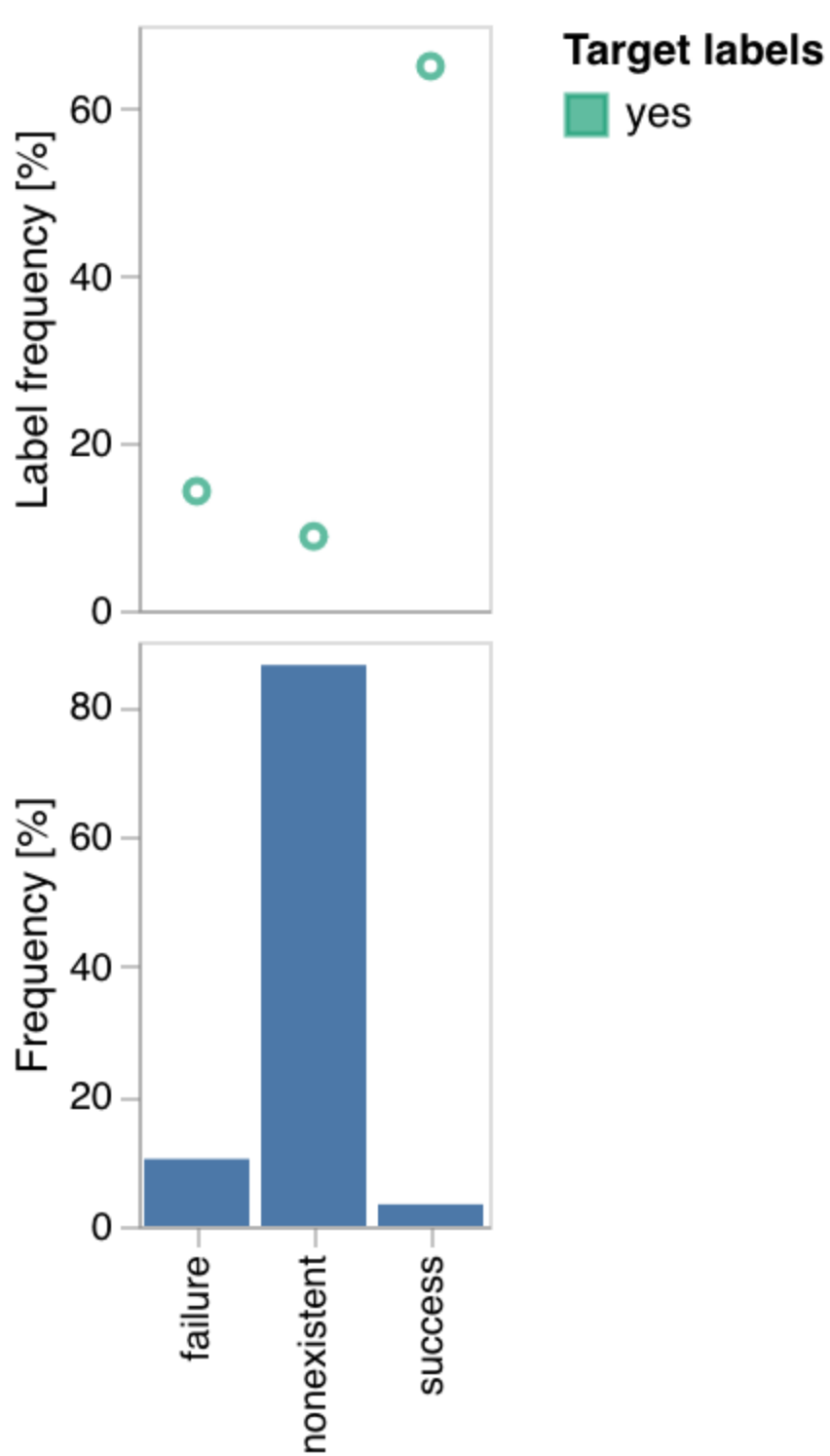
key	value	
Feature name	emp.var.rate	
Type	numeric	
Prediction power	0.0283	
Valid	100%	
Missing	0%	
Outliers	0%	
Min	-3.4	
Max	1.4	
Mean	0.0819	



Histogram of emp.var.rate with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency.

poutcome · categorical

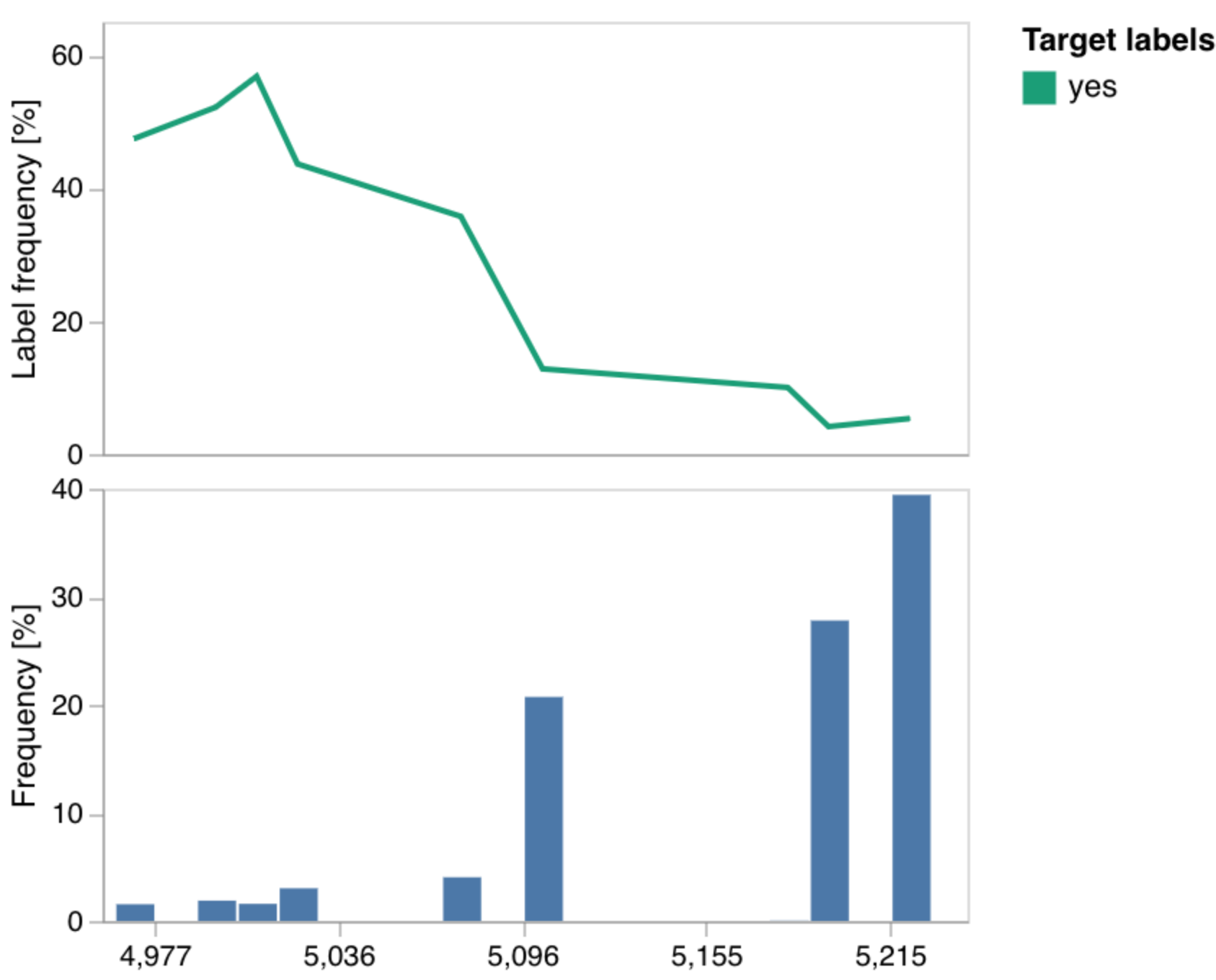
key	value	
Feature name	poutcome	
Type	categorical	
Prediction power	0.0189	
Valid	100%	
Missing	0%	



Histogram of the frequent values of poutcome with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency.

nr.employed · numeric

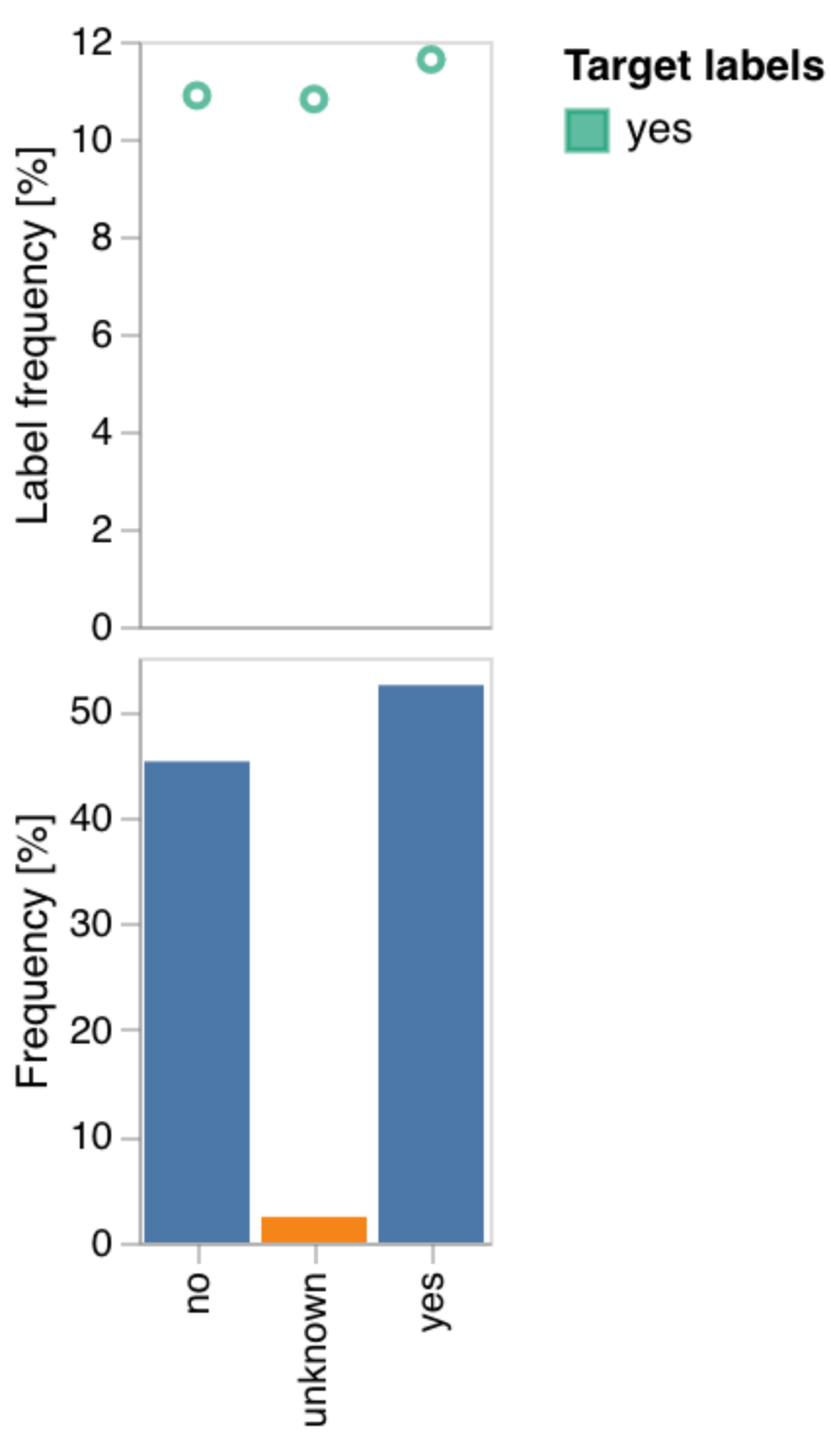
key	value	
Feature name	nr.employed	
Type	numeric	
Prediction power	0.0182	
Valid	100%	
Missing	0%	
Outliers	0%	
Min	4.96e+03	
Max	5.23e+03	
Mean	5.17e+03	



Histogram of nr.employed with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency.

housing · categorical

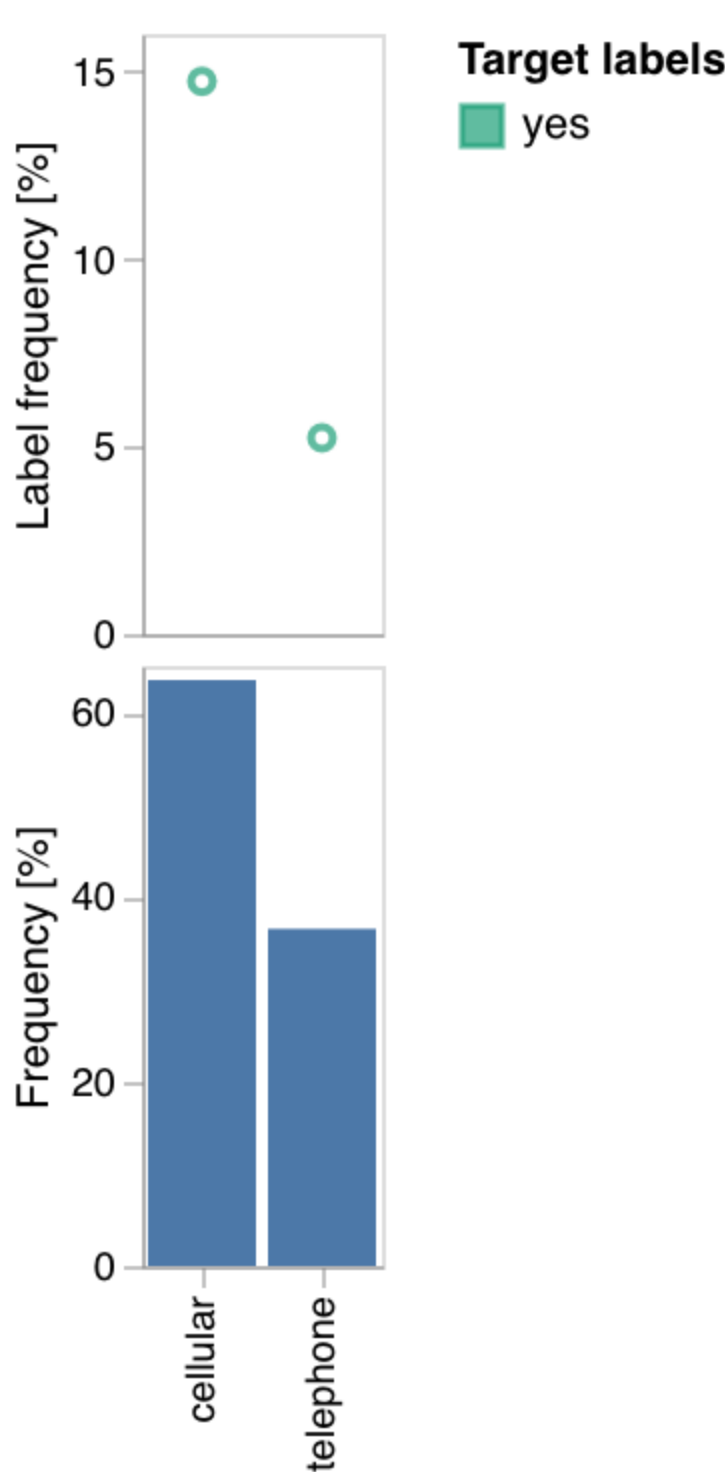
key	value	
Feature name	housing	
Type	categorical	
Prediction power	0.0174	
Valid	100%	
Missing	0%	



Histogram of the frequent values of housing with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency.

contact · binary

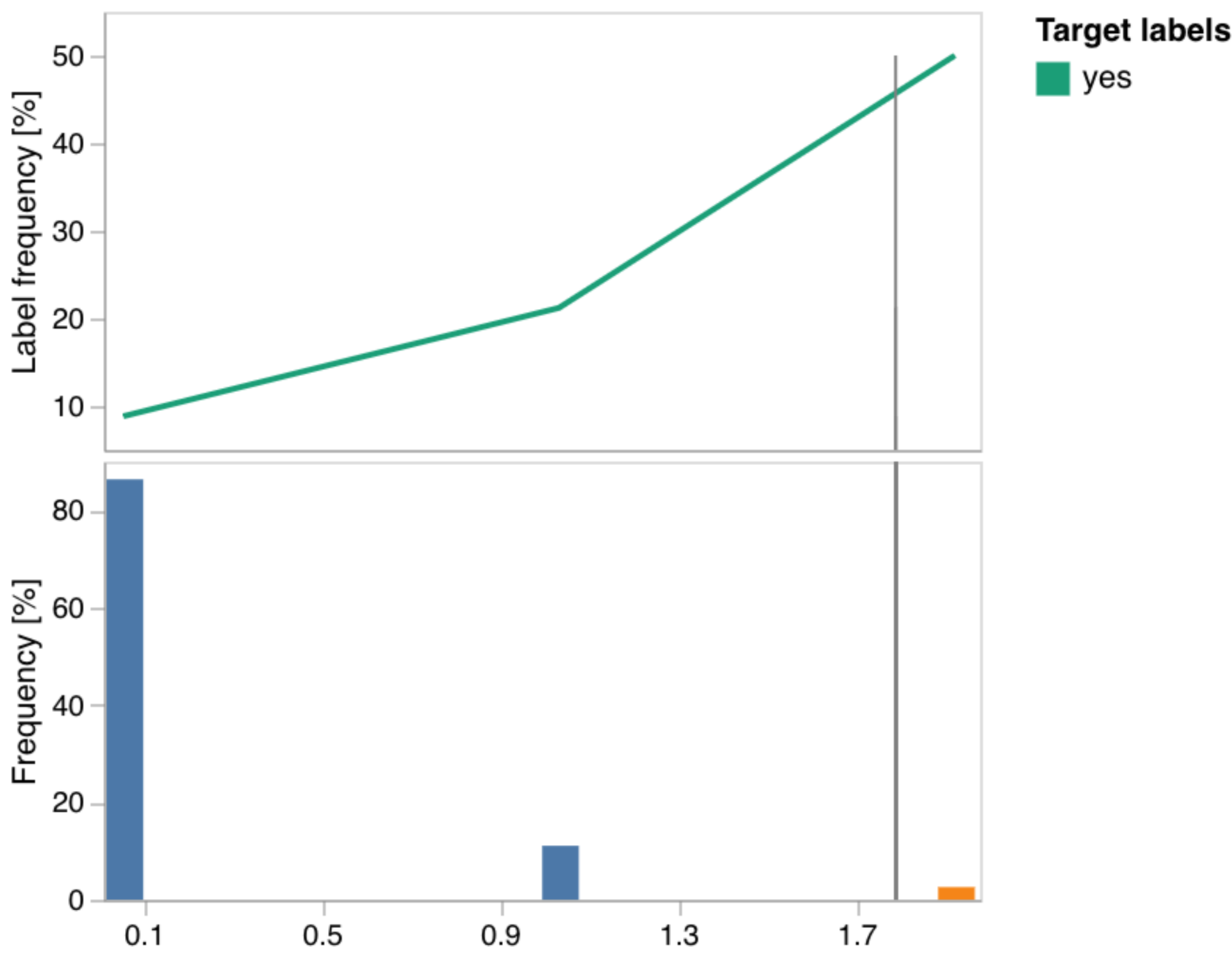
key	value	
Feature name	contact	
Type	binary	
Prediction power	0.0174	
Valid	100%	
Missing	0%	



Histogram of the frequent values of contact with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency.

previous · numeric

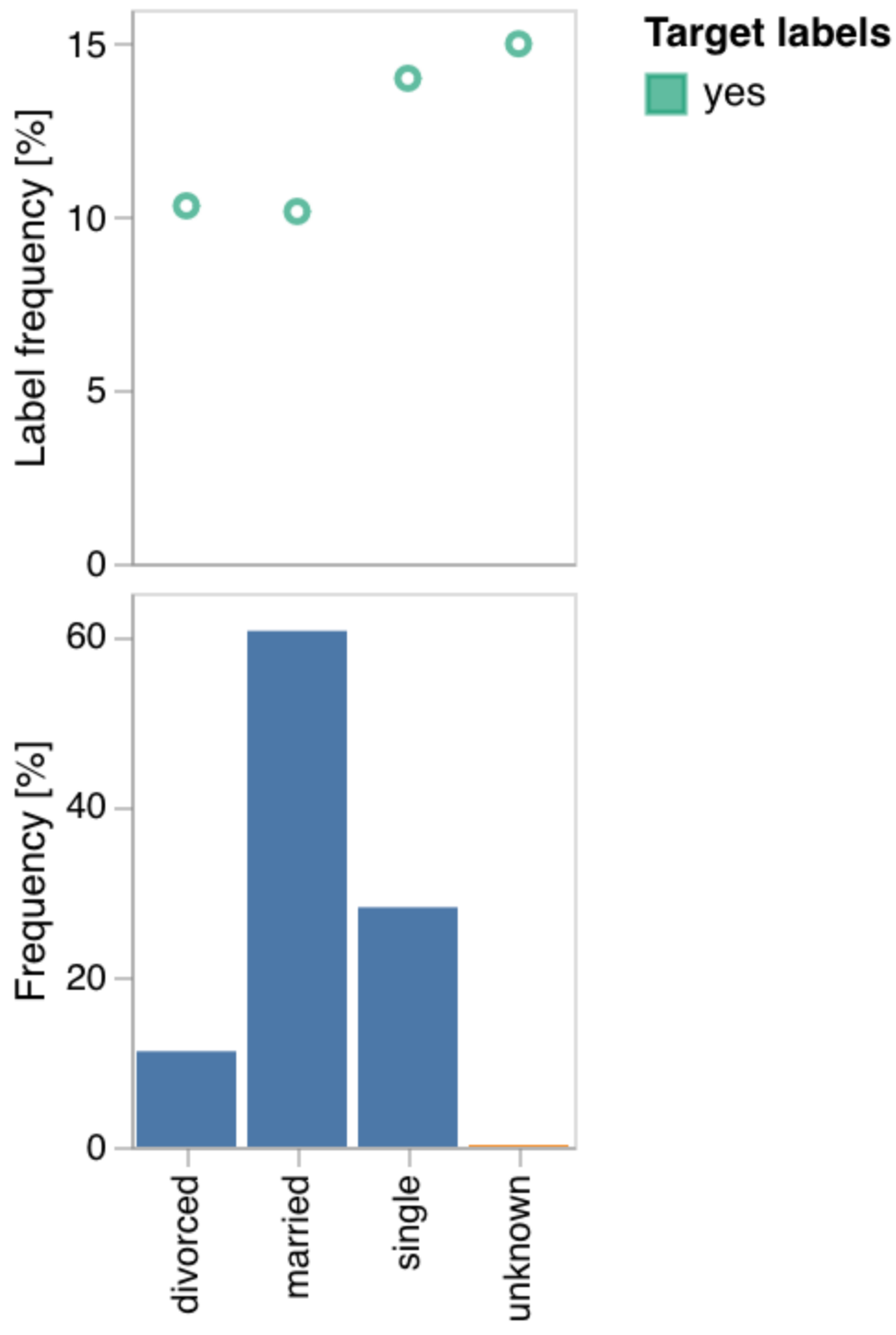
key	value	
Feature name	previous	
Type	numeric	
Prediction power	0.0167	
Valid	100%	
Missing	0%	
Outliers	2.58%	
Min	0	
Max	7	
Mean	0.173	



Histogram of previous with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency. The red bars contain outliers and the value below them is the outliers average.

marital · categorical

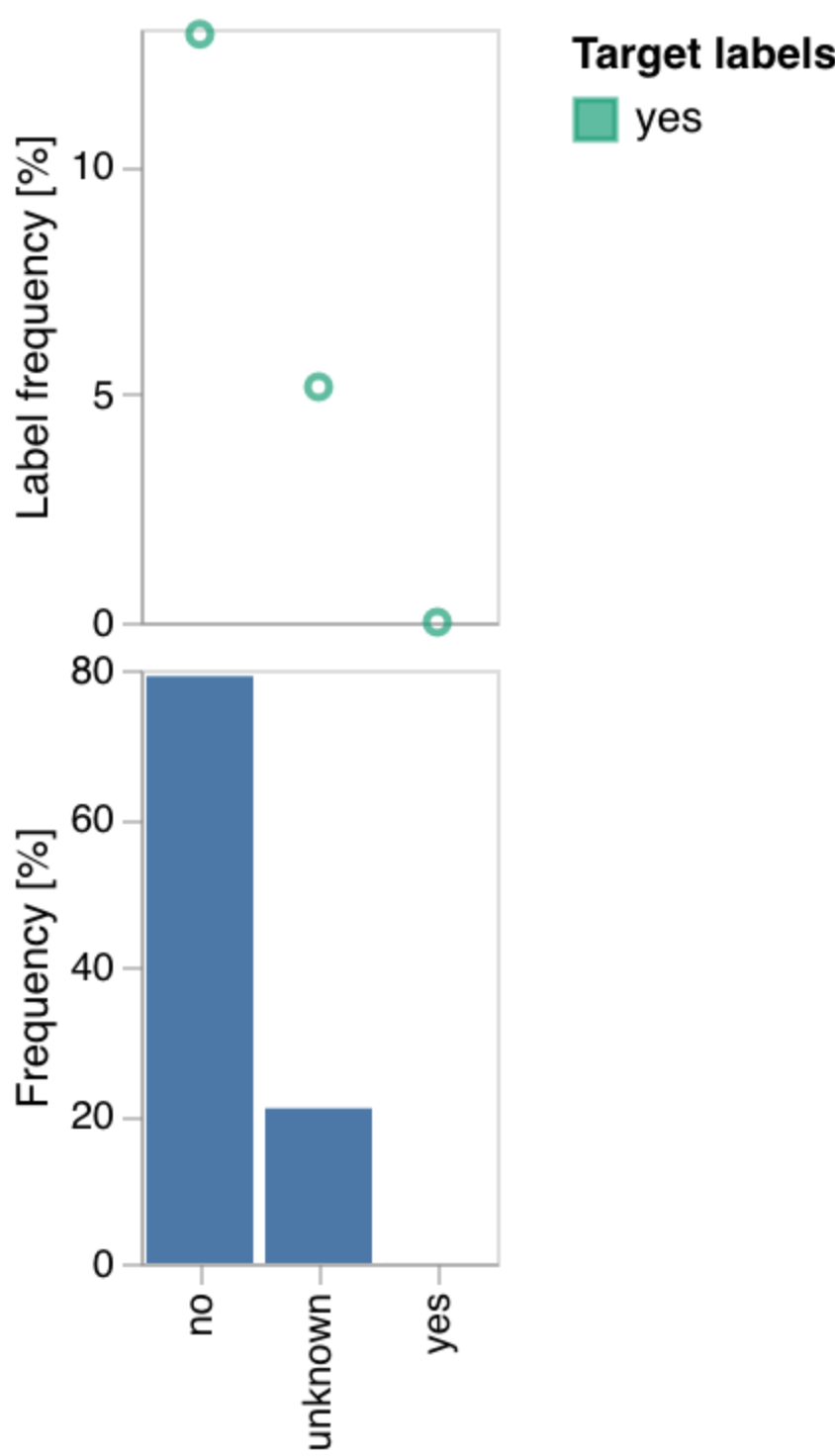
key	value	
Feature name	marital	
Type	categorical	
Prediction power	0.0153	
Valid	100%	
Missing	0%	



Histogram of the frequent values of marital with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency.

default · categorical

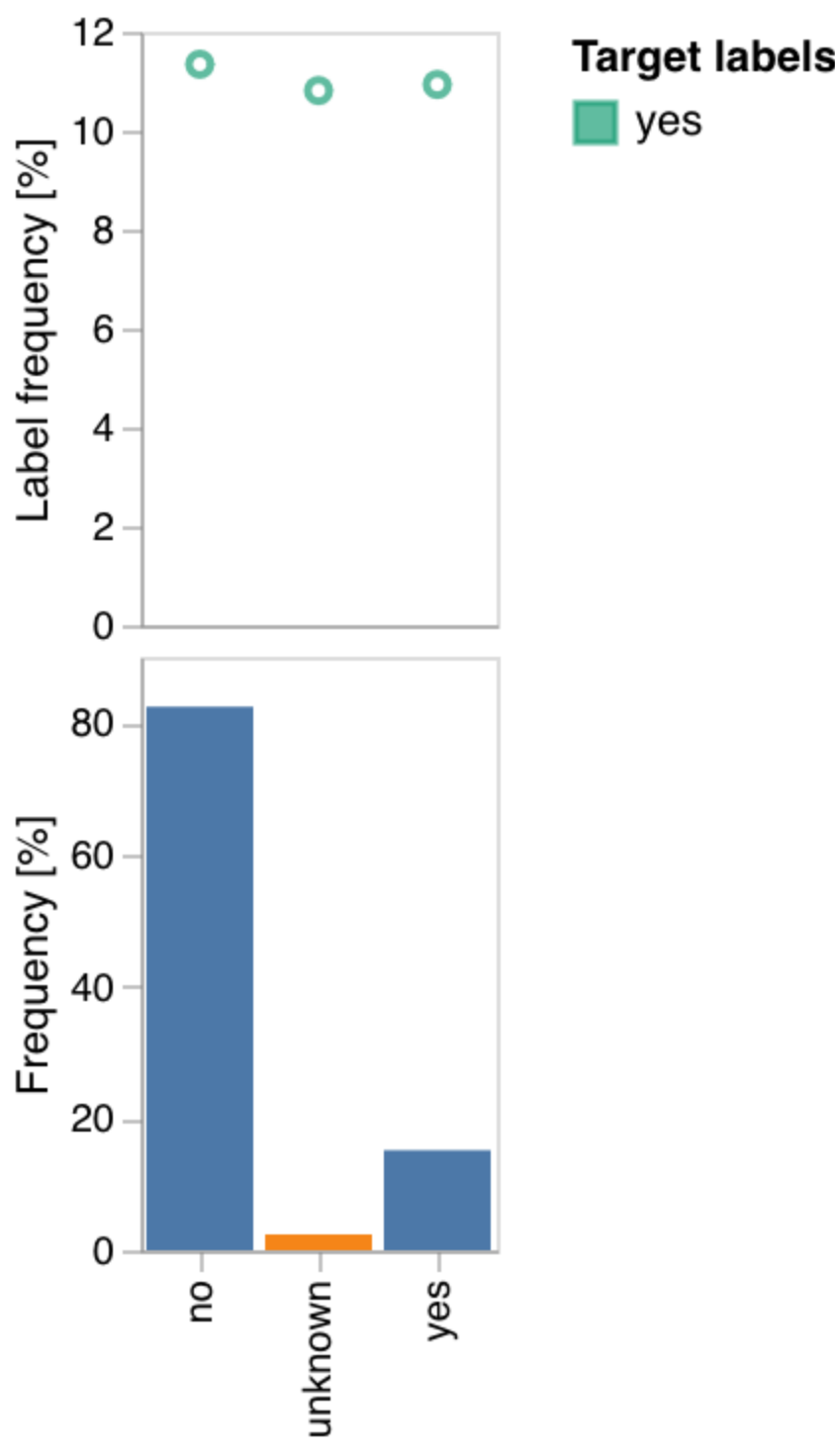
key	value	
Feature name	default	
Type	categorical	
Prediction power	0.00945	
Valid	100%	
Missing	0%	



Histogram of the frequent values of default with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency.

loan · categorical

key	value	
Feature name	loan	
Type	categorical	
Prediction power	0.00799	
Valid	100%	
Missing	0%	



Histogram of the frequent values of loan with the corresponding target distribution. The lower plot provides the feature distribution and the upper - the corresponding target class frequency.

Definitions

Feature types

Numeric: Numeric values, either floats or integers. For example: age, income. The machine learning models assume that numeric values are ordered and a distance is defined over them, that is 3 is closer to 4 than to 10 and $3 < 4 < 10$.

Categorical: The column entries belong to a set of unique values which is usually much smaller than the number of entries in the column. For example, a column of length 100 containing the unique values "Dog", "Cat" and "Mouse". The values could be numeric, text or combination of both e.g. "Horse", "House", 8, "Love" and 3.1 are all valid values and can be found in the same categorical column. As opposed to numeric features, the machine learning model does not assume order or distance on the values of a categorical feature (even when all the values are numbers).

Binary: A special case of categorical feature where the cardinality of the set of unique values is 2.

Text: A text column contains many non-numeric unique values. In extreme cases, all the elements of the column are unique i.e., no two entries are the same.

Datetime: This column contains date and/or time information.

Feature Statistics

Prediction power

Outliers (in numeric features and regression target): Outliers are detected using two statistics that are robust to outliers: median and robust standard deviation (RSTD). RSTD is derived by clipping the feature values to the range [5 percentile, 95 percentile] and calculating the standard deviation of the clipped vector. All values larger than $\text{median} + 5 * \text{RSTD}$ or smaller than $\text{median} - 5 * \text{RSTD}$ are considered to be outliers.

Skew (in numeric features and regression target): Skew measures the symmetry of the distribution and is defined as the third moment of the distribution divided by the third power of the standard deviation. The skewness of the normal distribution or any other symmetric distribution is zero. Positive values imply that the right tail of the distribution is longer than the left tail and negative values – vice versa. As a thumb rule, a distribution is considered skewed when the absolute value of the skew is larger than 3.

Kurtosis (in numeric features and regression target): Pearson's kurtosis measures the heaviness of the tail of the distribution and is defined as the fourth moment of the distribution divided by the square of the second moment. The kurtosis of the normal distribution is 3. Thus, kurtosis values lower than 3 imply that the distribution is concentrated around the mean and the tails are lighter than the tails of the normal distribution. On the other hand, Kurtosis values higher than 3 imply heavier tails and/or outliers.

Values

Numeric features / regression target: All values that could be casted to finite floats are valid. Missing values are not valid.

Categorical / binary / text features / classification target: All values that are not missing are valid.

Datetime features: All values that could be casted to datetime object are valid. Missing values are not valid.

Invalid values: Either missing or could not be casted to the desired type. See the definition of valid values for more information

Quick model metrics

Regression

R2 (coefficient of determination): R2 is the proportion of the variation in the target that is predicted by the model. R2 is in the range of [-infy, 1] where 1 is the score of the model that predicts the target perfectly and 0 is the score of the trivial model that always predicts the target mean.

MSE: Mean squared error. MSE is in the range [0, infy] where 0 is the score of the model that predicts the target perfectly

MAE: Mean absolute error. MAE is in the range [0, infy] where 0 is the score of the model that predicts the target perfectly

RMSE: Root mean square error. RMSE is in the range [0, infy] where 0 is the score of the model that predicts the target perfectly

Median absolute error: In the range [0, infy] where 0 is the score of the model that predicts the target perfectly

Classification

Accuracy: The ratio of samples that are predicted accurately. Accuracy is in the range [0, 1] where 0 is the score of the model that predicts all samples wrong and 1 is the score of the perfect model.

Balanced accuracy: The ratio of samples that are predicted accurately when class weights are adjusted to balance the data. That is, all classes are given the same importance, regardless of their frequency. Balanced accuracy is in the range [0, 1] where 0 is the score of the model that predicts all samples wrong and 1 is the score of the perfect model.

ROC-AUC: (Binary classification) Area under the receiver operating characteristic curve. ROC-AUC is in the range [0, 1] where a random model will yield a score of 0.5 and the perfect model 1.

ROC-AUC (OVR): (Multi-class classification) Area under the receiver operating characteristic curve calculated separately for each label using one vs rest and the average is reported. ROC-AUC is in the range [0, 1] where a random model will yield a score of 0.5 and the perfect model 1.

Precision: Precision is defined for a specific class. Precision is the fraction of true positives out of all the instances that the model classified as that class. Precision is in the range [0, 1] where 1 is the score of the model that have no false positives for the class. For binary classification, we report the precision of the positive class.

Recall: Recall is defined for a specific class. Recall is the fraction of the relevant class instances that are successfully retrieved. Recall is in the range [0, 1] where 1 is the score of the model that classifies correctly all the instances of the class. For binary classification, we report the recall of the positive class.

F1: F1 is defined for a specific class and it is the harmonic mean of the precision and recall. F1 is in the range [0, 1] where 1 is the score of the perfect model. For binary classification, we report F1 for the positive class.

Pattern Learning

Textual Patterns

Patterns describe the textual format of a string using an easy to read format. The following are examples of textual patterns:

- "{digits:4-7}" describes a sequence of digits of length between 4 and 7.
- "{alnum:5}" describes an alpha-numeric string of length exactly 5.

Canvas infers the patterns by looking at samples of non-empty strings from your data. It can describe many of the commonly used patterns. The relevance (%) indicates how much of the data is estimated to match the pattern. Using the textual pattern, you can see which rows in your data you need to correct or drop.

The following describes the patterns that Canvas can recognize:

Pattern	Textual Format
{alnum}	alphanumeric strings
{any}	any string of word characters
{digits}	a sequence of digits
{lower}	a lowercase word
{mixed}	a mixed-case word
{name}	a word beginning with a capital letter
{upper}	an uppercase word
{whitespace}	whitespace characters

A word character is either an underscore or a character that may appear in a word in any language. For example, the strings ‘Hello_world’ and ‘écoute’ both consist of word characters. ‘H’ and ‘é’ are both examples of word characters.